

Module 06: Learning — Parameter and Structure Learning

Course: Active Inference 101 | **Unit:** Mathematical Frameworks | **Audience:** First-semester undergraduates

Learning Objectives

1. Define **parameter learning** mathematically: updating the sufficient statistics of the A, B, C, D matrices.
2. Introduce **Bayesian Model Reduction** (BMR) as the mechanism for structure learning.
3. Explain how learning operates on a **slower timescale** than perception using Dirichlet distributions.

Introduction

Perception updates beliefs about hidden states (fast timescale). Learning updates the model itself — its parameters and structure (slow timescale). This module formalizes the difference mathematically.

Key Concepts

1. Parameter Learning — Updating Model Parameters

In the POMDP, the A and B matrices are initially uncertain. **Parameter learning** means updating them based on experience.

Mathematically, we represent uncertainty about parameters using **Dirichlet distributions**:

- Before experience: The A matrix starts with weak beliefs — A_prior = small concentration parameters (e.g., $\alpha = 1$)
- After experience: Each time the agent observes state-observation pair (s, o), the corresponding entry is incremented: $\alpha(o, s) \rightarrow \alpha(o, s) + 1$

- Over time: The entries grow, the distribution sharpens, and the A matrix becomes more certain

This is **Dirichlet-Categorical conjugacy** — a mathematically elegant property where the posterior stays in the same distribution family as the prior.

2. Concentration Parameters and Confidence

The concentration parameters α control how confident the agent is:

- **Small α (e.g., [1, 1, 1]):** Maximum uncertainty — all outcomes equally likely
- **Large α (e.g., [100, 5, 5]):** High confidence — heavily favors the first outcome
- **Sum of α :** Total "experience count" — larger sum = stronger confidence = slower updating

This explains why young animals learn fast (small α , weak priors) while experienced adults change slowly (large α , strong priors).

3. Learning Rate Emerges from Experience

The effective learning rate decreases naturally:

- After 1 observation: Learning rate $\approx 1/2$ (each new data point has 50% influence)
- After 100 observations: Learning rate $\approx 1/101$ (each new data point has $\sim 1\%$ influence)
- After 1000 observations: Learning rate $\approx 1/1001$ ($\sim 0.1\%$ influence)

This automatic decay of learning rate is optimal — early observations should matter more, later observations refine but don't fundamentally change the model.

4. Structure Learning — Bayesian Model Reduction (BMR)

Structure learning asks: "Does my model have the right structure?" — too many states? Wrong factorization? Unnecessary connections?

Bayesian Model Reduction compares the current model to simpler alternatives:

- Compute the model evidence F for the current model

- Compute F for simpler models (with some connections removed)
- If a simpler model has equal or better evidence $\rightarrow$ adopt it

BMR prunes unnecessary model complexity, implementing the mathematical version of Occam's razor. It's done offline (during rest/sleep) and is computationally cheap once the full model is learned.

5. Learning vs. Inference — Timescale Separation

Property	Inference (Perception)	Learning
What updates?	$q(s)$ — beliefs about states	Parameters α — the model itself
Timescale	Milliseconds	Minutes to years
Mathematical object	Posterior over states	Posterior over parameters
Effect of one observation	Can completely change beliefs	Slight nudge to parameters

Summary

Parameter learning updates the A and B matrices using Dirichlet distributions, with learning rate naturally decreasing with experience. Structure learning (BMR) prunes unnecessary model complexity by comparing simpler alternatives. Learning operates on a slower timescale than perception, with the timescale separation emerging naturally from the accumulation of concentration parameters.

Further Reading

- Friston, K. J. et al. (2016). Active inference and learning. *Neuroscience & Biobehavioral Reviews*, 68, 862-879.
- Friston, K. J. et al. (2018). Bayesian model reduction. *arXiv:1805.07092*.
- Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active Inference*. MIT Press. (Chapter 7)